DO NOW, AO1 REVISION

Induced potential, transformers and the National Grid

Magnetism and electromagnetism
Paper 2, Separate Physics only
22 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State what is induced by the generator effect.**	A potential difference (and a current, in a complete circuit).	☐
2 ★	**State what is needed for the generator effect to occur.**	A conductor moving relative to a magnetic field, or a changing magnetic field around a conductor.	☐
3 ★	**Give three ways to increase the size of the induced potential difference.**	Use a stronger magnetic field, move faster, or use more turns on the coil.	☐
4	**Describe how to reverse the direction of the induced potential difference.**	Reverse the direction of movement or of the magnetic field.	☐
5 ★	**State the difference between an alternator and a dynamo.**	An alternator generates alternating current; a dynamo generates direct current.	☐
6	**Name the effect used in microphones.**	The generator effect	☐
7 ★	**Describe how a microphone works.**	Sound makes a diaphragm and coil vibrate in a magnetic field, inducing a changing potential difference.	☐
8	**Name the effect used in transformers.**	The generator effect	☐
9 ★	**Describe the structure of a transformer.**	A primary coil and a secondary coil wound on an iron core.	☐
10	**Describe what a step-up transformer does.**	It increases the potential difference.	☐
11	**Describe what a step-down transformer does.**	It decreases the potential difference.	☐
12 ★	**Compare the numbers of turns on the coils of a step-up transformer.**	More turns on the secondary coil than on the primary.	☐

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	QUESTION	ANSWER	✓
13	**Compare the numbers of turns on the coils of a step-down transformer.**	Fewer turns on the secondary coil than on the primary.	☐
14 ★	**State how power input and power output are related in an ideal transformer.**	They are equal.	☐
15	**Name the type of current supplied to the primary coil of a transformer.**	Alternating current	☐
16	**Name the type of current induced in the secondary coil of a transformer.**	Alternating current	☐
17 ★	**Explain why a transformer needs an iron core.**	Iron is easily magnetised, so it carries the changing magnetic field from the primary coil to the secondary coil.	☐
18 ★	**State what the magnetic field of an induced current does.**	It opposes the change that caused it.	☐
19 ★	**Explain why a transformer does not work with a direct current.**	Direct current gives a constant magnetic field; induction needs a changing field.	☐
20	**State the equation linking the potential differences and numbers of turns of a transformer.**	p.d. across primary ÷ p.d. across secondary = turns on primary ÷ turns on secondary ($V_p \div V_s = n_p \div n_s$)	☐
21 ★	**Describe the potential difference–time graph for an alternator.**	It alternates: a wave going equally above and below zero.	☐
22 ★	**Describe the potential difference–time graph for a dynamo.**	It rises and falls but never goes below zero.	☐

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